

Global Fit Program

Moritz Lang

December 2022

1 Instrument Response

1.1 Gaussian Laserpuls with pulselength Δt

$$g(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \left(\frac{t-t_0}{\sigma}\right)^2\right) \quad (1)$$

$$g(t) = \frac{2}{\Delta t} \sqrt{\frac{\ln 2}{\pi}} \exp\left(-4 \ln 2 \left(\frac{t-t_0}{\Delta t}\right)^2\right) \quad (2)$$

2 Fit Functions

2.1 Exponential decay

$$f_n(t, A_n, \tau_n) = A_n \cdot \exp\left(-\frac{t}{\tau_n}\right) \quad (3)$$

3 Full Model

For fit function, convolute model with response function:

$$F(t, \Delta t, t_0, A_1, \tau_1) = g * f_n(t) \quad (4)$$

4 Obtaining DAS

Assume the decay can be fitted with a sum of functions according to Eq. 5

$$F(t) = \sum_n F_n(t, *Params) \quad (5)$$

*Params are the corresponding parameters for the Fit functions and the instruments response.

$$D = \begin{pmatrix} F_1(t, *Params_1) \\ F_2(t, *Params_2) \\ F_3(t, *Params_3) \\ \dots \end{pmatrix}$$

Measured times $\rightarrow$

$$A = \begin{pmatrix} \text{Spektrum1} & \left| \right. & \text{Spektrum2} & \left| \right. & \text{Spektrum3} & \left| \right. & \dots \end{pmatrix}$$

Measured times $\rightarrow$

$$F = \begin{pmatrix} \text{DAS1} & \left| \right. & \text{DAS2} & \left| \right. & \text{DAS3} & \left| \right. & \dots \end{pmatrix}$$

$$A = F \cdot D$$

$$A \cdot D^T = F \cdot D \cdot D^T$$

$$F = A \cdot D^T \cdot (D \cdot D^T)^{-1}$$